14.1 Ring axioms

A nonempty set R, equipped with addition and multiplication, is a **ring** if:

1. $(R, +)$ is an abelian group. Addition is closed, associative and commutative, has identity 0, and gives every element an additive inverse.
2. Multiplication is closed and associative.
3. Both distributive laws hold, $a(b + c) = ab + ac$ and $(a + b)c = ac + bc$.

Following the convention for this chapter, a general ring need not have commutative multiplication or a multiplicative identity. A **ring with identity** has an element 1; a **commutative ring** also satisfies $ab = ba$. Assumptions such as commutativity and the existence of 1 will be stated when discussing modular arithmetic, units, integral domains, fields and CRT. Properties of these special rings cannot silently be used for all rings.

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The zero ring and ideals

If a ring with identity satisfies $0 = 1$, then every $a = a1 = a0 = 0$. It has just one element and is the **zero ring**. Integral domains and fields exclude it.

The distinction between a subring and an ideal concerns absorption under multiplication. Containing 0 and being closed is not enough to make a subset an ideal.

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14.2 Examples of rings

- $\mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$ are commutative rings with identity.
- $M_n(\mathbb{R})$, with ordinary matrix addition and multiplication, is a ring with identity. For $n \geq 2$, multiplication is generally not commutative, so steps requiring commutativity need justification.
- For a positive integer n, the residue classes in $\mathbb{Z}_n$ form a ring under addition and multiplication followed by reduction modulo n.
- Polynomial rings $F[x]$ will be used later; their coefficients come from a field or ring.

Ring elements need not be numbers. Functions also form rings under pointwise addition and multiplication.

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Two explicit counterexamples in a matrix ring

Let

$$A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad B = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}.$$

Direct multiplication gives

$$AB = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \qquad BA = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}.$$

Thus $AB \neq BA$. Also $A \neq 0$ but $A^2 = 0$, so A is a nonzero zero divisor. In general, $AX = AY$ does not allow cancellation of A. Matrix multiplication still satisfies associativity and both distributive laws. Noncommutativity does not prevent a structure from being a ring.

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A ring whose elements are sets

For a finite set U, take the subsets of U as elements, symmetric difference $\triangle$ as addition, and intersection $\cap$ as multiplication. The empty set is the additive identity. Since $A \triangle A = \varnothing$, each A is its own additive inverse. The multiplicative identity is U.

For each $u \in U$, use a bit to record membership. Symmetric difference becomes bitwise XOR, and intersection becomes bitwise AND. Each ring axiom holds coordinate by coordinate by the corresponding axiom in $\mathbb{Z}_2$. This is a commutative ring. If $|U| = k$, it has 2^k elements and is isomorphic to $(\mathbb{Z}_2)^k$.

If $k \geq 2$, two distinct singletons A, B satisfy $A \cap B = \varnothing$. They are nonzero zero divisors, so the ring is not a field. Yet every element satisfies $A^2 = A$, the **idempotent** property. Whether division is possible depends on the operations, not on whether the elements look like numbers, sets, matrices or functions.

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14.3 Basic consequences of the axioms

For all $a, b \in R$,

$$a0 = 0 = 0a, \qquad a(-b) = -(ab) = (-a)b.$$

Indeed, $a0 = a(0+0) = a0 + a0$; cancellation in the additive group gives $a0 = 0$. The argument uses distributivity and additive inverses, rather than assuming the familiar integer rule.

The additive identity is unique, because $0 + 0'$ equals both 0 and $0'$. If b, c are additive inverses of a, then

$$b = b + (a + c) = (b + a) + c = c.$$

Thus additive inverses are unique. Since $ab + a(-b) = a(b - b) = 0$, uniqueness gives $a(-b) = -(ab)$. The other side is analogous, and $(-a)(-b) = ab$ follows. None of these results requires commutative multiplication.

If $ab = 0$ with **both** $a \neq 0$ and $b \neq 0$, then a, b are nonzero zero divisors. Knowing only $b \neq 0$, while allowing $a = 0$, does not establish that b is a zero divisor.

In $\mathbb{Z}_6$, $2 \cdot 3 = 0$, so multiplicative cancellation does not hold universally.

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14.4 Units, zero divisors and integral domains

In a commutative ring with identity, u is a **unit** if some v satisfies $uv = 1$. The units form a multiplicative group $R^\times$. In a noncommutative ring, both $uv = 1$ and $vu = 1$ are required. An **integral domain** is a nonzero commutative ring with identity satisfying

$$ab = 0 \implies a = 0 \text{ or } b = 0.$$

The integers form an integral domain. For composite $n > 1$, $\mathbb{Z}_n$ does not.

Proof of the prime-modulus criterion. If p is prime and $ab \equiv 0 \pmod{p}$, Euclid's lemma gives $p \mid a$ or $p \mid b$. Conversely, if $n = rs$ with $1 < r, s < n$, the classes of r, s are nonzero and have product zero.

An integral domain need not be a field. In $\mathbb{Z}$, only $1, -1$ have multiplicative inverses, although other nonzero elements exist.

A **field** is a nonzero commutative ring with identity in which every nonzero element is a unit. Division by nonzero elements is available in a field. A general ring guarantees additive inverses, but not multiplicative inverses.

Cancellation and finite integral domains

In a commutative ring, $a \neq 0$ and $ab = ac$ yield only $a(b - c) = 0$. If a is not a zero divisor, this implies $b = c$. Conversely, cancellation for a particular nonzero a forces $b = 0$ whenever $ab = 0 = a0$. Thus cancellation by a is equivalent to a not being a zero divisor.

In a nonzero commutative ring with identity, a nonzero unit cannot be a zero divisor. Multiplying $ab = 0$ by a^{-1} gives $b = 0$. Every field is therefore an integral domain. The converse fails for $\mathbb{Z}$, where 2 is not invertible.

Every finite integral domain is a field. For $a \neq 0$, the map $m_a : x \mapsto ax$ is injective by cancellation. An injection from a finite set to itself is surjective, so $1 = ax$ for some x. Every nonzero a is invertible. Finiteness is essential. On $\mathbb{Z}$, $x \mapsto 2x$ is injective but not surjective.

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The unit group and characteristic

In a ring with identity, v is the multiplicative inverse of u if $uv = vu = 1$. In a commutative ring, one equality suffices. The inverse is unique, since $v = v(uw) = (vu)w = w$. Products of units are units, with

$$(uv)^{-1} = v^{-1}u^{-1}.$$

Consequently all units form a group. In a noncommutative ring, the order of these two inverse factors cannot be reversed without justification.

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Characteristic of a ring

The **characteristic** is the least positive integer p such that $p \cdot 1 = 0$, or 0 if no such integer exists. An integral domain has $1 \neq 0$, so its characteristic is not 1. If a positive characteristic were composite, $p = ab$ with $1 < a, b < p$, then

$$(a \cdot 1)(b \cdot 1) = 0.$$

Minimality of p makes both factors nonzero, a contradiction. A positive characteristic of an integral domain must therefore be prime. In a finite integral domain, the sequence $0, 1, 2 \cdot 1, \ldots$ repeats, so its characteristic is a prime.

For example, $\mathbb{Z}_{12}$ has characteristic 12 and is not an integral domain. The characteristic of $\mathbb{Q}$ is 0. Characteristic measures repeated addition of the identity, rather than the numeral used to name it.

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14.5 Subrings and ideals

A nonempty subset $S \subseteq R$ is a **subring** if it is itself a ring under the restricted operations. Our convention does not require S to contain 1_R. The exact test is

$$a, b \in S \implies a - b \in S \text{ and } ab \in S.$$

For sufficiency, taking $a = b$ gives $0 \in S$, then $0 - a$ gives additive inverses and $a - (-b)$ gives additive closure. Associativity and distributivity are inherited. Necessity follows directly from the subring axioms.

A **unital subring** must additionally contain 1_R. For example, $2\mathbb{Z}$ is a subring and ideal of $\mathbb{Z}$, but does not contain 1 and has no multiplicative identity of its own under the inherited multiplication. Do not change subring conventions midway through an argument.

A **two-sided ideal** I is an additive subgroup satisfying $ri, ir \in I$ for all $r \in R, i \in I$. This is stronger than multiplication being closed only within I. If an ideal in a ring with identity contains 1, then $r = r1 \in I$ for every r, so $I = R$.

Define operations on cosets by

$$(a + I) + (b + I) = (a + b) + I, \qquad (a + I)(b + I) = ab + I.$$

14.5 Subrings and ideals (continued)

Replacing representatives by $a + i, b + j$ changes the product by $aj + ib + ij$. Two-sided absorption puts every term in I; the change in the sum also lies in I. Both operations are well defined, producing the **quotient ring** R/I. In a noncommutative ring, ib cannot arbitrarily become bi. An arbitrary subring lacks the absorption needed to descend multiplication to cosets.

For instance, the constants $\mathbb{Z} \subseteq \mathbb{Z}[x]$ form a subring but not an ideal, since multiplying an integer by x need not give a constant.

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Finite subrings and the limits of ideal closure

A nonempty finite subset S closed under addition and multiplication is already a subring. For $a \in S$, some positive multiples satisfy $ia = ja$, with $i < j$. Additive cancellation in R yields $(j - i)a = 0$. This positive multiple lies in S, so $0 \in S$, and

$$-a = (j - i - 1)a \in S.$$

If the coefficient is zero, use the zero element just established. Additive inverses and closure make S an additive subgroup.

Finiteness cannot be omitted. Nonnegative integers are closed under ordinary addition and multiplication and contain $0, 1$, but have no additive inverse for 1. They are not a subring. Also, $\mathbb{Z}$ is a unital subring of $\mathbb{Q}$, but not an ideal because $\frac{1}{2} \cdot 1 \notin \mathbb{Z}$.

A nonzero commutative ring with identity having exactly the ideals $\{0\}, R$ is a field. For $a \neq 0$, the principal ideal $(a) = \{ra : r \in R\}$ is nonzero, hence equals R; thus $1 = ra$ and a is invertible. Conversely, any nonzero ideal of a field contains $a \neq 0$; multiplying by a^{-1} puts 1 in the ideal, so it is the whole field. The nonzero hypothesis keeps the two ideals distinct.

Ideals of the integers and of residue rings

Every ideal $I \subseteq \mathbb{Z}$ equals $m\mathbb{Z}$ for a unique $m \geq 0$. If $I = \{0\}$, take $m = 0$. Otherwise take the least positive element m of I. For $a \in I$, division gives $a = qm + r$, $0 \leq r < m$. Since $a, qm \in I$, also $r \in I$. Minimality forces $r = 0$, so $I \subseteq m\mathbb{Z}$. Absorption gives the reverse inclusion.

For an ideal J of $\mathbb{Z}_n$, its preimage

$$I = \{a \in \mathbb{Z} : [a]_n \in J\}$$

is an ideal of $\mathbb{Z}$ containing $n\mathbb{Z}$. Thus $I = d\mathbb{Z}$ with $d \mid n$. Conversely each positive divisor $d \mid n$ gives $J = ([d])$. This is a bijection between ideals of $\mathbb{Z}_n$ and positive divisors of n. Such J has n/d elements; its quotient has d elements and is isomorphic to $\mathbb{Z}_d$.

For example, in $\mathbb{Z}_{12}$, $([4]) = \{0, 4, 8\}$. The quotient has four classes,

$$\{0, 4, 8\},\ \{1, 5, 9\},\ \{2, 6, 10\},\ \{3, 7, 11\}.$$

Sending each class to its residue modulo 4 is a bijection preserving both operations, so $\mathbb{Z}_{12}/([4]) \cong \mathbb{Z}_4$.

14.6 Operations in $\mathbb{Z}/n\mathbb{Z}$

Fix a positive integer n. Representatives a and $a + kn$ must determine the same class. The quotient operations are well defined because

$$(a + kn) + (b + \ell n) \equiv a + b \pmod{n},$$
$$(a + kn)(b + \ell n) \equiv ab \pmod{n}.$$

Every extra term is a multiple of n. Moreover,

$$[a] \text{ is a unit} \iff \gcd(a, n) = 1.$$

Proof. If the gcd is 1, Bézout gives $sa + tn = 1$, hence $[s][a] = [1]$. Conversely, if $ab + kn = 1$, any common divisor of a, n divides 1.

The additive identity is $[0]$ and the multiplicative identity is $[1]$. For $n = 1$, these are the same class, producing the zero ring.

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Complete classification of units and zero divisors

Let $n > 1$ and $[a] \neq [0]$, and put $d = \gcd(a, n)$. If $d = 1$, $[a]$ is a unit. If $d > 1$, set $b = n/d$. Then $1 \leq b < n$ and $n \mid ab$, so $[b] \neq [0]$ and $[a][b] = 0$. This explicitly constructs a zero-divisor witness. A unit cannot have zero product with a nonzero class. Thus every nonzero class belongs to exactly one of these two categories.

Euler's function $\varphi(n)$ counts the residue classes coprime to n. If the distinct prime factors are $p_1, \ldots, p_r$, inclusion–exclusion on their multiples among $0, \ldots, n-1$ gives

$$\varphi(n) = n \prod_{p \mid n} \left(1 - \frac{1}{p}\right).$$

There are $n - 1 - \varphi(n)$ nonzero zero divisors. For $n = 18$,

$$\varphi(18) = 18(1 - 1/2)(1 - 1/3) = 6.$$

The units are $1, 5, 7, 11, 13, 17$; the other 11 nonzero classes are zero divisors. For example, $6 \cdot 3 \equiv 0$ and $5 \cdot 11 \equiv 1$.

For $n = 1$, the only class is both 0 and 1, and is invertible by the definition of a unit. This ring is neither a field nor an integral domain. The counting formula above is explicitly restricted to $n > 1$.

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Linear congruences require a solvability test

To solve $ax \equiv b \pmod{n}$, with $n \geq 1$, put $d = \gcd(a, n)$. A solution implies $n \mid ax - b$, hence $d \mid b$. Conversely, if $d \mid b$, division by d gives

$$\frac{a}{d}x \equiv \frac{b}{d} \pmod{n/d}.$$

The new coefficient is coprime to the new modulus, so there is a unique x_0 modulo n/d. All solutions modulo n are

$$x = x_0 + k(n/d), \qquad k = 0, \ldots, d-1,$$

which are d distinct classes. If $n/d = 1$, the reduced equation is just $0 \equiv 0$, and all classes modulo n solve the original equation; no ordinary division needs to be defined.

For $6x \equiv 8 \pmod{14}$, $d = 2$ divides 8. Reduction gives $3x \equiv 4 \pmod{7}$. The inverse of 3 is 5, so $x \equiv 20 \equiv 6 \pmod{7}$. The two solutions modulo 14 are $6, 13$. Substitution gives $36 \equiv 8$ and $78 \equiv 8$. Replacing the right side by 9 gives no solution, because $2 \nmid 9$.

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14.7 Modular inverses and Euler's theorem

Let a be an integer and $n > 1$. If $\gcd(a, n) = 1$, the extended Euclidean algorithm produces s with $sa \equiv 1 \pmod{n}$. Euler's theorem states that

$$a^{\varphi(n)} \equiv 1 \pmod{n}$$

for a coprime to n. Fermat's little theorem is the prime-modulus case, where $n = p$ is prime and $p \nmid a$:

$$a^{p-1} \equiv 1 \pmod{p}.$$

If $\gcd(a, n) > 1$ and $n > 1$, then a has no multiplicative inverse modulo n. A power congruence happening to have a solution does not make a available for division.

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Euler's theorem by permutation and cancellation

Let $n > 1$, with units $u_1, \ldots, u_\varphi$. Since $[a]$ is a unit, multiplication by a permutes the units. Indeed, $au_i = au_j$ implies $u_i = u_j$ after multiplying by a^{-1}; finiteness gives a bijection. Commutativity then gives

$$a^{\varphi(n)} u_1 \cdots u_\varphi = u_1 \cdots u_\varphi.$$

The product on the right is a unit and can be cancelled by its inverse, proving the theorem. The permutation and cancellation fail without coprimality. For example, $2^{\varphi(8)} = 2^4 \equiv 0 \pmod 8$, not 1. For prime p, $\varphi(p) = p - 1$, yielding Fermat's little theorem. The version also allowing $p \mid a$ is $a^p \equiv a$; division by a is not justified in that case.

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Binary modular exponentiation, bit by bit

To compute $a^e \bmod n$ for $n > 1, e \geq 0$, initialize $r = 1$ and scan all binary bits of e from left to right. For each bit b, first set $r \leftarrow r^2 \bmod n$, then, if $b = 1$, set $r \leftarrow ra \bmod n$. After reading a prefix of value q, the invariant is $r \equiv a^q$. Appending b changes the exponent to $2q + b$, proving the update. For $e = 0$, return $1 \bmod n$.

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Binary modular exponentiation: 5^{13} mod 17

For $a = 5, e = 13 = (1101)_2, n = 17$:

Bit read	Prefix q	After squaring modulo 17	After multiplying by 5 if needed
Initial	0	No squaring	1
1	1	1	5
1	3	8	6
0	6	2	2
1	13	4	3

Thus $5^{13} \equiv 3 \pmod{17}$. An implementation initialized at $r = a$ must skip the leading significant bit; starting at a and scanning every bit double-counts it. The algorithm uses $O(\log(e+1))$ modular multiplications. This is not the same number of bit operations; each large-integer modular multiplication has its own cost.

14.8 Chinese Remainder Theorem

For pairwise coprime positive moduli $m_1, \ldots, m_k$, the system

$$x \equiv a_i \pmod{m_i}$$

has exactly one solution modulo $N = \prod_i m_i$. To construct it, let $N_i = N/m_i$. Choose t_i with $t_i N_i \equiv 1 \pmod{m_i}$, possible since $\gcd(N_i, m_i) = 1$. Then

$$x = \sum_i a_i t_i N_i$$

satisfies every congruence. For uniqueness, the difference of two solutions is divisible by every m_i, hence by N because the moduli are pairwise coprime.

For the example $x \equiv 2 \pmod{3}$, $x \equiv 3 \pmod{5}$, write $x = 2 + 3t$. Then $2 + 3t \equiv 3 \pmod{5}$, so $3t \equiv 1 \pmod{5}$. Since $3^{-1} \equiv 2 \pmod{5}$, $t \equiv 2 \pmod{5}$, giving $x \equiv 8 \pmod{15}$.

Pairwise coprimality is sufficient for this form of CRT. More general versions use ideals.

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CRT as an isomorphism of ring representations

For pairwise coprime moduli greater than 1, define

$$\Theta : \mathbb{Z}_N \longrightarrow \prod_i \mathbb{Z}_{m_i}, \qquad [x]_N \longmapsto ([x]_{m_1}, \ldots, [x]_{m_k}).$$

It is well defined since each $m_i \mid N$, and it preserves addition and multiplication coordinatewise. CRT existence and uniqueness state exactly that it is bijective. Thus it is a ring isomorphism. Product-ring operations are coordinatewise operations, not decimal operations on the entries of a tuple.

For $N = 12 = 3 \cdot 4$, let $e_1 = 4, e_2 = 9$. Their images are $(1, 0), (0, 1)$, and

$$e_1^2 = e_1, \quad e_2^2 = e_2, \quad e_1 e_2 = 0, \quad e_1 + e_2 = 1 \pmod{12}.$$

Hence $\Theta^{-1}(a, b) = 4a + 9b \pmod{12}$.

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A complete CRT correspondence

Each cell reconstructs one residue modulo 12. The outlined cells are units.

CRT: each (a mod 3, b mod 4) determines one x mod 12

	b=0	b=1	b=2	b=3
a=0	x=0	x=9	x=6	x=3
a=1	x=4	**x=1**	x=10	**x=7**
a=2	x=8	**x=5**	x=2	**x=11**

Double border: units

x=1, 5, 7, 11

$4 \mapsto (1,0)$

$9 \mapsto (0,1)$

Each cell has x = 4a+9b mod 12. All 12 residues occur once; operations are coordinatewise.

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Compatibility when moduli are not coprime

For positive integers m, n and $x \equiv a \pmod{m}, x \equiv b \pmod{n}$, put $g = \gcd(m, n)$. A solution requires $g \mid (b - a)$. If that condition holds, writing $x = a + mk$ reduces the system to

$$(m/g)k \equiv (b - a)/g \pmod{n/g}.$$

The reduced coefficient and modulus are coprime. Thus k is unique modulo n/g, and x is unique modulo $\operatorname{lcm}(m, n)$. This also proves sufficiency of the compatibility condition. Uniqueness is not generally modulo mn.

For $x \equiv 5 \pmod{8}, x \equiv 9 \pmod{12}$, $g = 4$ divides the difference 4. Set $x = 5 + 8k$. Then $2k \equiv 1 \pmod{3}$, so $k \equiv 2 \pmod{3}$, and $x \equiv 21 \pmod{24}$. Indeed, $21 \bmod 8 = 5$ and $21 \bmod 12 = 9$. Replacing the second residue by 10 makes the difference 5, not divisible by 4, so there is no solution.

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Deterministic applications of modular arithmetic

A linear congruential generator uses $x_{t+1} = (ax_t + c) \bmod m$. Its finite state space guarantees eventual repetition, but not period m. For $a = 5, c = 1, m = 16, x_0 = 0$, the actual sequence is

$$0 \to 1 \to 6 \to 15 \to 12 \to 13 \to 2 \to 11 \to 8 \to 9 \to 14 \to 7 \to 4 \to 5 \to 10 \to 3 \to 0.$$

This seed visits all 16 states before returning, so the period is 16. Changing only a to 6 gives $0 \to 1 \to 7 \to 11 \to 3 \to 3$, with a transient followed by a cycle of length 1. Determinism and finite periodicity imply neither independent uniform random values nor cryptographic security.

The hash function $h(k) = k \bmod 13$ maps integers to 13 buckets. The keys $13, 26, 39$ all map to 0; more than 13 distinct keys force a collision. The remainder specifies the bucket, while collision handling and distribution quality are separate issues.

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When an affine encoding can be decoded uniquely

On $\mathbb{Z}_n$, $E(x) = ax + b$ is a permutation if and only if $\gcd(a, n) = 1$. In the coprime case,

$$D(y) = a^{-1}(y - b)$$

is its inverse. If $d = \gcd(a, n) > 1$, the distinct classes x and $x + n/d$ have the same image, so E is not injective.

For $n = 26, a = 5, b = 7$, $5^{-1} = 21$, $E(4) = 27 \equiv 1$, and $D(1) = 21(1 - 7) = -126 \equiv 4$. With $a = 6$, both $x = 0$ and $x = 13$ map to 7; unique decoding is impossible. This is an example of algebraic invertibility, not a claim that a simple affine cipher provides secure encryption.

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14.9 Ring homomorphisms

A **ring homomorphism** $\varphi : R \to S$ satisfies, for every $a, b \in R$,

$$\varphi(a+b) = \varphi(a) + \varphi(b), \qquad \varphi(ab) = \varphi(a)\varphi(b).$$

Following our convention for general rings, preservation of 1 is not assumed. A **unital ring homomorphism** additionally satisfies $\varphi(1_R) = 1_S$. A map preserving only addition and multiplication may send 1 to another idempotent, so domain units need not become codomain units.

For a bijective homomorphism, the inverse preserves both operations. Write $x = \varphi(a), y = \varphi(b)$; then $\varphi^{-1}(x+y) = a+b$, and similarly for multiplication. A bijective homomorphism is a **ring isomorphism**. Equal cardinalities guarantee only a set bijection, not preservation of operations.

For a field F and a fixed $a \in F$, evaluation $F[x] \to F$, $f \mapsto f(a)$, is a unital ring homomorphism.

What a homomorphism preserves

From $\varphi(0) = \varphi(0) + \varphi(0)$, additive cancellation gives $\varphi(0) = 0$. Then $\varphi(a) + \varphi(-a) = \varphi(0)$ yields $\varphi(-a) = -\varphi(a)$. Images of additive subgroups are additive subgroups, and images of subrings are subrings.

If I is an ideal of S, then $\varphi^{-1}(I)$ is closed under subtraction. For i in this preimage and $r \in R$, $\varphi(ri) = \varphi(r)\varphi(i) \in I$ and likewise $\varphi(ir) \in I$. Thus the preimage is an ideal of R. Conversely, the image of an ideal of R is generally only an ideal of $\operatorname{Im} \varphi$. If φ is surjective onto S, any $s \in S$ is $\varphi(r)$, which proves absorption by all of S.

For the inclusion $\mathbb{Z} \to \mathbb{Q}$, the ideal $2\mathbb{Z}$ has the same image, but this is not an ideal of $\mathbb{Q}$, since $\frac{1}{2} \cdot 2 = 1 \notin 2\mathbb{Z}$. Surjectivity is essential in that direction.

If φ is surjective and R has 1, every $s = \varphi(r)$ satisfies $\varphi(1)s = s = s\varphi(1)$. Thus $\varphi(1)$ is the multiplicative identity of S. In the surjective case, identity preservation follows from the axioms.

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14.10 Kernel, image and the first isomorphism theorem

The **kernel**

$$\ker \varphi = \{r \in R : \varphi(r) = 0_S\}$$

is an ideal, and the image is a subring. Moreover, φ is injective if and only if $\ker \varphi = \{0\}$. The first isomorphism theorem states

$$R/\ker \varphi \cong \operatorname{Im} \varphi.$$

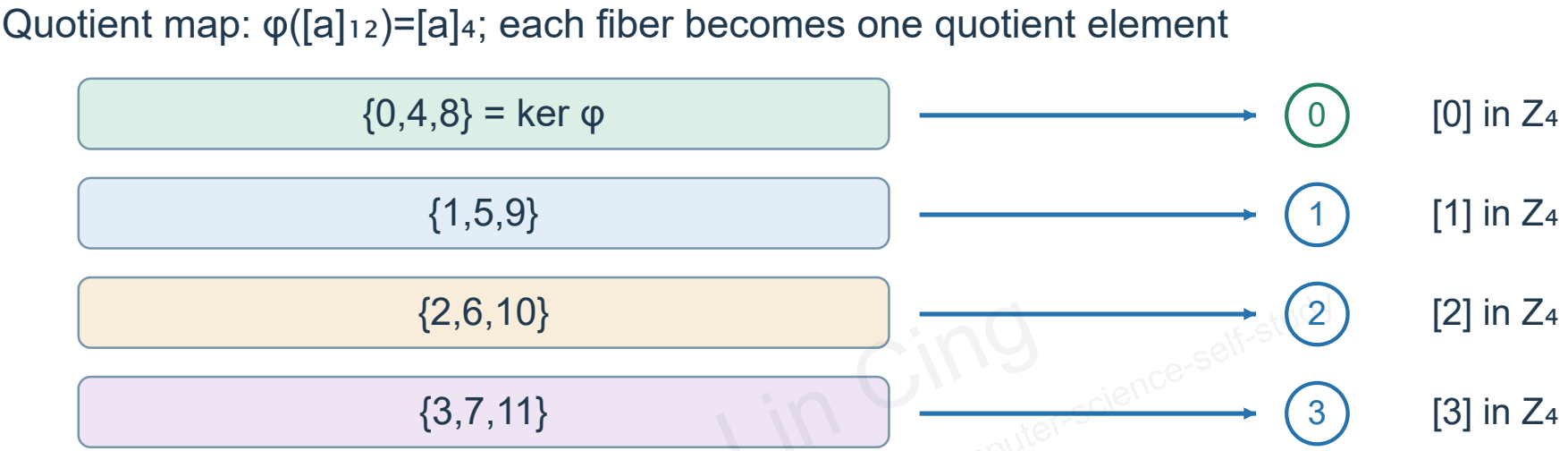
For $\varphi : \mathbb{Z} \to \mathbb{Z}_n$, $k \mapsto [k]$, the kernel is $n\mathbb{Z}$, giving $\mathbb{Z}/n\mathbb{Z} \cong \mathbb{Z}_n$.

Proof. The map $r + \ker \varphi \mapsto \varphi(r)$ is well defined because representatives of one coset differ by a kernel element. It preserves both operations and surjects onto the image. Equal outputs mean the representatives differ by a kernel element, hence their cosets coincide, proving injectivity. This constructs the isomorphism rather than merely asserting a symbolic equality.

The kernel is an ideal by the preceding preimage result with $I = \{0\}$. If φ is injective, $\varphi(r) = 0 = \varphi(0)$ forces $r = 0$. Conversely, if the kernel is zero and $\varphi(a) = \varphi(b)$, then $\varphi(a - b) = 0$ forces $a = b$. Both directions of the injectivity test are proved.

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Four fibers of a quotient map



The natural map $\varphi([a]_{12}) = [a]_4$ is well defined because a difference divisible by 12 is also divisible by 4. Its kernel is $\{0, 4, 8\}$. The three elements in each color group are distinct in $\mathbb{Z}_{12}$ but have one output. The quotient identifies each fiber as a single element; it does not treat the original elements in that fiber as separate quotient elements.

Changing the modulus requires well-definedness

The candidate $[a]_m \mapsto [a]_n$ is well defined if and only if $n \mid m$. Necessity follows because $[0]_m = [m]_m$ must have equal images. For sufficiency, if m divides the difference between representatives, then so does n. Once well defined, the map preserves addition, multiplication and 1.

Thus $\mathbb{Z}_{12} \to \mathbb{Z}_4$ is valid, but the same formula from $\mathbb{Z}_4$ to $\mathbb{Z}_{12}$ is not: $[0]_4 = [4]_4$ would have distinct images. Choosing the least nonnegative representative defines a set function but does not repair additivity. The image of $[3]_4 + [1]_4 = [0]_4$ is 0, whereas $3 + 1 = 4$ in $\mathbb{Z}_{12}$.

A valid nonunital example is

$$\psi : \mathbb{Z}_4 \to \mathbb{Z}_6, \qquad [a] \mapsto [3a].$$

It preserves addition. Multiplication is preserved since $9ab \equiv 3ab \pmod 6$, and $4 \cdot 3 \equiv 0$ ensures independence from representatives. The image $\{0, 3\}$ has its own multiplicative identity 3, rather than 1 of $\mathbb{Z}_6$. Its kernel is $\{0, 2\}$, so $\mathbb{Z}_4/\{0, 2\} \cong \{0, 3\} \cong \mathbb{Z}_2$.

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14.11 Boundaries and common errors

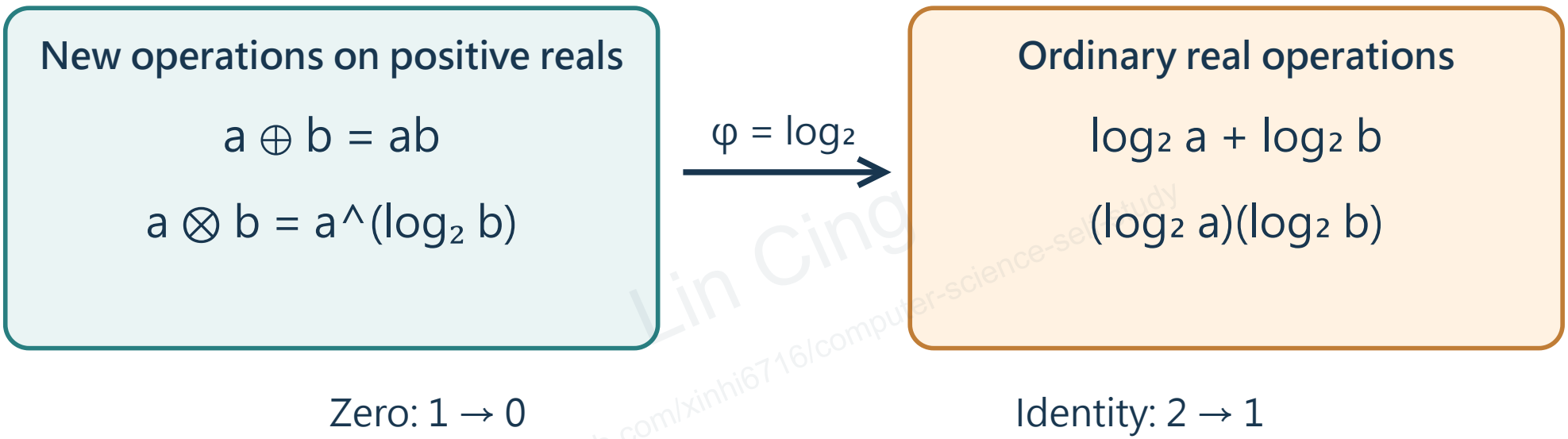
- $(\mathbb{Z}_n, +, \cdot)$ is a ring, but nonzero classes need not be units.
- Products of nonzero elements can be zero for composite moduli.
- $\mathbb{Z} \subset \mathbb{Q}$ is a unital subring but not an ideal; $2\mathbb{Z} \subset \mathbb{Z}$ is an ideal lacking 1.
- A kernel must contain zero and absorb multiplication by the ambient ring.
- The pairwise gcd conditions must be checked before applying the coprime form of CRT.

These structures recur in RSA in Chapter 16 and finite fields in Chapter 17. A field requires additional axioms.

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New operations on the positive reals

On $S = \mathbb{R}_{>0}$, define $a \oplus b = ab$ and $a \otimes b = a^{\log_2 b}$. Both outputs are positive and remain in S.



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Compare these operations with ordinary real arithmetic using $\phi(a) = \log_2 a$. The symbols $\oplus, \otimes$ are newly defined operations.

Why is the logarithm a ring isomorphism?

The map ϕ is a bijection from positive reals to reals, with inverse $t \mapsto 2^t$. For all $a, b > 0$,

$$\phi(a \oplus b) = \log_2(ab) = \phi(a) + \phi(b), \qquad \phi(a \otimes b) = \log_2(a^{\log_2 b}) = \phi(a)\phi(b).$$

Thus ϕ preserves both operations and is an isomorphism to the ordinary real ring. Ring axioms transfer through this bijection: for example, associativity holds after applying ϕ, and injectivity gives equality in S.

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Zero, identity, and inverses

Under the inverse map, real 0 and 1 correspond to 1 and 2. Hence the new additive zero is 1, and the new multiplicative identity is 2.

Every $a > 0$ has additive inverse $1/a$, because $a \oplus (1/a) = 1$. The multiplicative units are exactly $a \neq 1$, with inverse

$$a^{\otimes -1} = 2^{1/\log_2 a}, \qquad a \otimes a^{\otimes -1} = 2.$$

When $a = 1$, $\phi(a) = 0$; division by $\log_2 a$ is undefined and no multiplicative inverse exists.

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Example: 2 and 8 under both operations

$$2 \oplus 8 = 16, \qquad 2 \otimes 8 = 2^{\log_2 8} = 8.$$

Here $\phi(2) = 1$ and $\phi(8) = 3$, while $\phi(16) = 4 = 1 + 3$ and $\phi(8) = 3 = 1 \cdot 3$. The same inputs give different results under the two operations, so the operator matters.

For the same element 8, its additive inverse is $1/8$, since $8 \oplus (1/8) = 1$. Its multiplicative inverse is $2^{1/3}$, since $8 \otimes 2^{1/3} = 8^{1/3} = 2$. These results use the new zero and identity.

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Transporting operations changes zero and one

On $\mathbb{Q}$, define

$$a \oplus b = a + b - 2, \qquad a \otimes b = (a - 2)(b - 2) + 2.$$

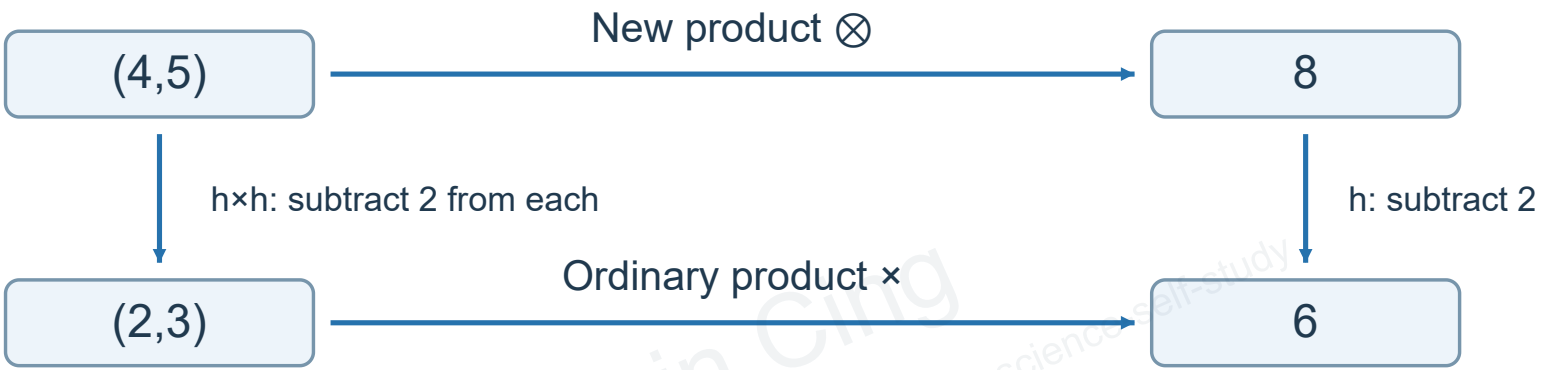
The bijection $h(a) = a - 2$ satisfies $h(a \oplus b) = h(a) + h(b)$ and $h(a \otimes b) = h(a)h(b)$. It transforms both new operations into ordinary rational operations. Transporting the ring axioms through h gives a field isomorphic to $\mathbb{Q}$.

The new additive zero is $h^{-1}(0) = 2$, and the new multiplicative identity is $h^{-1}(1) = 3$. The additive inverse of a is $4 - a$. For $a \neq 2$, its new multiplicative inverse is $2 + 1/(a - 2)$. For example, the inverse of 4 is $5/2$, and $4 \otimes (5/2) = 3$, the new identity. Requiring this product to equal the ordinary numeral 1 would use the wrong operation structure.

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Transporting operations changes zero and one (continued)

Transported multiplication: $4 \otimes 5 = 8$ corresponds to $2 \times 3 = 6$



New product then h, or h on each input then ordinary product: both paths give 6.

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Checking an isomorphism

First establish a bijection, then verify preservation of addition and multiplication. For transported operations, determine zero, one, and inverses using the new operations.

References and attribution

Lin Cing, handwritten discrete mathematics notes. Ralph P. Grimaldi, Discrete and Combinatorial Mathematics, 5th ed., Ch. 14.

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